

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
MEC136	ENGINEERING GRAPHICS AND CAD	2	2	0	4	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50					

Course Outcomes :Through this course students should be able to

CO1 :: recognize the fundamentals of engineering drawing and AutoCAD tool.

CO2 :: understand the conceptual framework of orthographic projections with an acquaintance of 2D AutoCAD commands.

CO3 :: understand the drawing of orthographic projections and 2D sketches on AutoCAD.

CO4 :: practice the technique to draw the sectional views on drawing sheet and AutoCAD.

CO5 :: build isometric views on drawing sheet and 3D models on AutoCAD.

CO6 :: develop surfaces of various objects and create 3D models on AutoCAD.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	ENGINEERING DRAWING WITH AN INTRODUCTION TO AUTOCAD	DHANANJAY JOLHE	MC GRAW HILL

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	ENGINEERING DRAWING	N.D. BHAT & M. PANCHAL	CHAROTAR PUBLISHING HOUSE PVT. LTD.
R-2	ENGINEERING GRAPHICS	K C JOHN	PRENTICE HALL
R-3	ENGINEERING DRAWING	M.B. SHAH & B.C. RANA	PEARSON

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=1TN19J44GK4&list=PL3OVqiyXzMVKmnOgnrmsVFu5RoVSvoJA_	Introduction to Engineering Drawing
AV-2	https://www.youtube.com/watch?v=0Q6QwvtjVm8	Drawing instruments
AV-3	https://www.youtube.com/watch?v=XS0lJsmy-qg&t=7s	Dimensioning

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AV-4	https://www.youtube.com/watch?v=onJIaSAkiEs&t=32s	Letter writing
AV-5	https://www.youtube.com/watch?v=_VktSY6SIH0&list=PL3OVqiyXzMVKmnOgnrmsVFu5RoVSvoJA_&index=3	Scales
AV-6	https://www.youtube.com/watch?v=JAxkYvp7Z6w&list=PL3OVqiyXzMVKmnOgnrmsVFu5RoVSvoJA_&index=2	Scales
AV-7	https://www.youtube.com/watch?v=TRh1OfZzzzA&list=PL3OVqiyXzMVKmnOgnrmsVFu5RoVSvoJA_&index=4	Scales
AV-8	https://www.youtube.com/watch?v=Rz4WWtDX2Kc&t=9s	Orthographic Projections
AV-9	https://www.youtube.com/watch?v=heIobpUVZqc	orthographic projections in first angle and third angle projections systems
AV-10	https://www.youtube.com/watch?v=mcxUTNkSyp4	First and third angle projections
AV-11	https://www.youtube.com/watch?v=cmR9cfWJRUI	AutoCAD
AV-12	https://www.youtube.com/watch?v=z1dHm9JKYYg	AutoCAD
AV-13	https://www.youtube.com/watch?v=rtkyNGEQXD4	AutoCAD
AV-14	https://www.youtube.com/watch?v=iCLGQNEAs7o	AutoCAD
AV-15	https://www.youtube.com/watch?v=0x4Mq3plkEw	Sectioning
AV-16	https://www.youtube.com/watch?v=mLy4xt4-Oqg	Sectioning
AV-17	https://www.youtube.com/watch?v=belZlq0ExQU	Sectioning
AV-18	https://www.youtube.com/watch?v=YZkPG7ywm14	Sectioning
AV-19	https://www.youtube.com/watch?v=WVhqUH4VD20	Sectioning
AV-20	https://www.youtube.com/watch?v=TEOcTMj8o7w	Sectioning
AV-21	https://www.youtube.com/watch?v=ClokZICFivQ	Sectioning
AV-22	https://www.youtube.com/watch?v=Wc9K8_VPRAA	Sectioning
AV-23	https://www.youtube.com/watch?v=glRMBJFygG0	Sectioning
AV-24	https://www.youtube.com/watch?v=UqJwADpprEU	Sectioning
AV-25	https://www.youtube.com/watch?v=viNCXvO9bzY	Isometric Views
AV-26	https://www.youtube.com/watch?v=LaKNBXZ4sfA	Isometric Views
AV-27	https://www.youtube.com/watch?v=8j7l1OWhMlE	Isometric Views
AV-28	https://www.youtube.com/watch?v=K_2uJPkCv0Y	Isometric Views
AV-29	https://www.youtube.com/watch?v=o3kA8yAw9So	Isometric Views
AV-30	https://www.youtube.com/watch?v=RQll_sGa2pc	Isometric Views
AV-31	https://www.youtube.com/watch?v=6reJsWxvGyM&list=PL7EE46D045BEB6CCF	Isometric Views
AV-32	https://www.youtube.com/watch?v=Z6TB8ZAPb9M	Isometric Views
AV-33	https://www.youtube.com/watch?v=Z6TB8ZAPb9M	Isometric Views

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AV-34	https://www.youtube.com/watch?v=6reJsWxvGyM	Isometric Views
AV-35	https://www.youtube.com/watch?v=KJUeY25QpVY&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr	Development of Surfaces
AV-36	https://www.youtube.com/watch?v=CIBq106nj7U&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr&index=2	Development of Surfaces
AV-37	https://www.youtube.com/watch?v=9-lwnNPG9wQ&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr&index=3	Development of Surfaces
AV-38	https://www.youtube.com/watch?v=idgLYAJ_8W4&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr&index=4	Development of Surfaces
AV-39	https://www.youtube.com/watch?v=nAhduVb-gXo&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr&index=5	Development of Surfaces
AV-40	https://www.youtube.com/watch?v=neTytXOMIE0&list=PLVdvOTOMfu5Q1obf9HirYCQOGtQC01SHr&index=6	Development of Surfaces

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Introduction to Engineering Drawing(line types)	T-1 R-1 R-2 R-3	AV-1 AV-2	Introduction to Engineering Drawing, instruments and types of lines	Student must be familiarized with the conceptual framework of engineering drawing	Presentations, white board demonstrations, AV aids	Industrial/ product drawings
		Introduction to Engineering Drawing(line types)	T-1 R-1 R-2 R-3	AV-1 AV-2	Introduction to Engineering Drawing, instruments and types of lines	Student must be familiarized with the conceptual framework of engineering drawing	Presentations, white board demonstrations, AV aids	Industrial/ product drawings

Week 1	Lecture 2	Introduction to Engineering Drawing(dimensioning)	T-1 R-1 R-2 R-3	AV-3	Introduction to dimensioning systems, methods and related concepts	Student must be acquainted with the dimensioning styles and techniques	Presentations, white board demonstrations, AV aids	Practice drawings, engineering drawings/ architectural drawings
Week 2	Lecture 3	Introduction to Engineering Drawing(single stroke vertical gothic letter writing)	T-1 R-1 R-2 R-3	AV-4	Basics of letter writing, significance, rules and related concepts	Student must know the method to draw the single stroke vertical gothic letters on drawing/ grid sheets	Presentations, white board demonstrations, AV aids	Industrial drawings
	Lecture 4	Introduction to Engineering Drawing(scales-plain and diagonal)	T-1 R-1 R-2 R-3	AV-5 AV-6 AV-7	Introduction to concept, usages, applications and types of scales in engineering	Student must understand the usage of scale for engineering drawing purposes and also be familiarized with the applications of their types	Presentations, white board demonstrations, AV aids	Layout drawing used by engineers while constructing of new house
Week 3	Lecture 5	Introduction to Engineering Drawing(scales-plain and diagonal)	T-1 R-1 R-2 R-3	AV-5 AV-6 AV-7	Introduction to concept, usages, applications and types of scales in engineering	Student must understand the usage of scale for engineering drawing purposes and also be familiarized with the applications of their types	Presentations, white board demonstrations, AV aids	Layout drawing used by engineers while constructing of new house
	Lecture 6	Introduction to Engineering Drawing(introduction to AutoCAD interface- units)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
		Introduction to Engineering Drawing(limits)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes

Week 3	Lecture 6	Introduction to Engineering Drawing(navigation)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
		Introduction to Engineering Drawing(OSNAP)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
		Introduction to Engineering Drawing(ortho)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
		Introduction to Engineering Drawing(UCS)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
		Introduction to Engineering Drawing(F-keys)	T-1	AV-11	Introduction of AutoCAD tool, applications, benefits, how to use commands for basic setup prior to creation of drawings	Student must understand the installation, preliminary setup followed by few basic commands prior to creation of 2D drawings	Live demonstration using AutoCAD tool followed by practice	Pictorial views of regular basic shapes
Week 4	Lecture 7	Projection of Points and Lines(Introduction)	T-1 R-1 R-2 R-3	AV-8	Introduction to orthographic projections, its industrial significance, rules and principles	Student must be familiarized with the concepts, needs, rules, techniques to draw the orthographic projections of objects	Presentations, white board demonstrations, AV aids	Actual objects, pictorial views of objects

Week 4	Lecture 7	Projection of Points and Lines(principles of quadrants and orthographic projections)	T-1 R-1 R-2 R-3	AV-8	Introduction to orthographic projections, its industrial significance, rules and principles	Student must be familiarized with the concepts, needs, rules, techniques to draw the orthographic projections of objects	Presentations, white board demonstrations, AV aids	Actual objects, pictorial views of objects
	Lecture 8	Projection of Points and Lines(concept of traces)	T-1 R-1 R-2 R-3	AV-8 AV-9	Introduction to first and third angle projection systems, their utilities along with practice	Student must comprehensively understand the differences between projection system and should be able to draw the orthographic views of points and lines in both first and third angle projection systems	Presentations, white board demonstrations, AV aids	Specific cases of points and lines
		Projection of Points and Lines(orthographic projection of points and lines (parallel, perpendicular and inclined to one plane))	T-1 R-1 R-2 R-3	AV-8 AV-9	Introduction to first and third angle projection systems, their utilities along with practice	Student must comprehensively understand the differences between projection system and should be able to draw the orthographic views of points and lines in both first and third angle projection systems	Presentations, white board demonstrations, AV aids	Specific cases of points and lines
Week 5	Lecture 9	Projection of Points and Lines(AutoCAD commands-line)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(circle)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes

Week 5	Lecture 9	Projection of Points and Lines(arc)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(polyline)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(and dimensioning style)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(hands-on-practice on AutoCAD)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(rectangle)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(polygon)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes
		Projection of Points and Lines(ellipse)	T-1 R-1 R-2 R-3	AV-10 AV-11 AV-12	Introduction to basic 2D drawing commands required for creation of orthographic views of basic simple shapes	Student must be able to select and use these commands for creation of simple orthographic drawings and operations	Live demonstration on AutoCAD software followed by practice	Pictorial/ 2D views of different objects/ shapes

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Week 5	Lecture 10	Orthographic Projections (Introduction)	T-1 R-1 R-2 R-3	AV-8 AV-9 AV-10	Describe the concept of orthographic views	Student should be able to understand and draw the orthographic views in both first and third angle projection systems	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
		Orthographic Projections (principle)	T-1 R-1 R-2 R-3	AV-8 AV-9 AV-10	Describe the concept of orthographic views	Student should be able to understand and draw the orthographic views in both first and third angle projection systems	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
		Orthographic Projections (orthographic projections in both first and third angle projections systems)	T-1 R-1 R-2 R-3	AV-8 AV-9 AV-10	Describe the concept of orthographic views	Student should be able to understand and draw the orthographic views in both first and third angle projection systems	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
Week 6	Lecture 11	Orthographic Projections (AutoCAD commands- linetype and its properties)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (move)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (rotate)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (trim)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software

Week 6	Lecture 11	Orthographic Projections (copy)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (erase)	T-1	AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
	Lecture 12	Orthographic Projections (practice)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (mirror)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (scale)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (fillet)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (chamfer)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (array)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software

Week 6	Lecture 12	Orthographic Projections (hands-on-practice on 2D drawings)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
Week 7	Lecture 13	Orthographic Projections (practice)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (mirror)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (scale)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (fillet)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (chamfer)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (array)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Orthographic Projections (hands-on-practice on 2D drawings)	T-1 R-1	AV-10 AV-11 AV-12	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software

		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						
Week 8	Lecture 15	Sectional Views (Introduction)	T-1 R-1 R-2 R-3	AV-15 AV-16	Describe the concept of sectional views, define what is meant by cutting plane, preparing different kind of sections, how to deal with ribs, webs, holes, spokes etc.	Student should be able to understand and show the internal/ hidden features of different objects	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
		Sectional Views(principle)	T-1 R-1 R-2 R-3	AV-15 AV-16	Describe the concept of sectional views, define what is meant by cutting plane, preparing different kind of sections, how to deal with ribs, webs, holes, spokes etc.	Student should be able to understand and show the internal/ hidden features of different objects	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
	Lecture 16	Sectional Views(sectional views (full, half, and offset) in first and third angle projection systems)	T-1 R-1 R-2 R-3	AV-17 AV-18 AV-19 AV-20 AV-21 AV-22 AV-23 AV-24	Describe the concept of sectional views, define what is meant by cutting plane, preparing different kind of sections, how to deal with ribs, webs, holes, spokes etc.	Student should be able to understand and show the internal/ hidden features of different objects	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
		Sectional Views(practice)	T-1 R-1 R-2 R-3	AV-17 AV-18 AV-19 AV-20 AV-21 AV-22 AV-23 AV-24	Describe the concept of sectional views, define what is meant by cutting plane, preparing different kind of sections, how to deal with ribs, webs, holes, spokes etc.	Student should be able to understand and show the internal/ hidden features of different objects	Presentations, white board demonstrations, AV aids	Industrial/ product drawings and pictorial views of objects
Week 9	Lecture 17	Sectional Views(AutoCAD commands- stretch)	T-1	AV-11 AV-12 AV-13	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software

Week 9	Lecture 17	Sectional Views(explode)	T-1	AV-11 AV-12 AV-13	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(offset)	T-1	AV-11 AV-12 AV-13	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(extend)	T-1	AV-11 AV-12 AV-13	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(join)	T-1	AV-11 AV-12 AV-13	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
	Lecture 18	Sectional Views(region)	T-1	AV-14	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(break)	T-1	AV-14	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(hatch)	T-1	AV-14	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
		Sectional Views(hatch-edit)	T-1	AV-14	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software

Week 9	Lecture 18	Sectional Views(hands-on-practice on 2D drawings)	T-1	AV-14	To discuss the details of using an AutoCAD software for the creation of 2D drawings and the pertinent settings	Student must be able to draw the 2D drawings using basic drawing and modify commands on AutoCAD tool	Practice on AutoCAD software	Software
Week 10	Lecture 19	Isometric Views (Introduction)	T-1 R-1 R-2 R-3	AV-25 AV-26 AV-27	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Isometric Views (terminology)	T-1 R-1 R-2 R-3	AV-25 AV-26 AV-27	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Isometric Views(isometric scale)	T-1 R-1 R-2 R-3	AV-25 AV-26 AV-27	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
	Lecture 20	Isometric Views(isometric views of prisms)	T-1 R-1 R-2 R-3	AV-28 AV-29 AV-30 AV-31	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Isometric Views(pyramids)	T-1 R-1 R-2 R-3	AV-28 AV-29 AV-30 AV-31	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
Week 11	Lecture 21	Isometric Views(and one object on other)	T-1 R-1 R-2 R-3	AV-30 AV-31 AV-32 AV-33 AV-34	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects

Week 11	Lecture 21	Isometric Views (dimensioning)	T-1 R-1 R-2 R-3	AV-30 AV-31 AV-32 AV-33 AV-34	Introduction to conceptual frameworks of the isometric projections with their importance, use, and practice	Student must use the methods to draw the different isometric shapes following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
	Lecture 22	Isometric Views(AutoCAD commands for 3D- 3P UCS rotation)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
		Isometric Views(standard shapes)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
		Isometric Views(extrude)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
		Isometric Views(revolve)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
		Isometric Views(presspull)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software

Week 11	Lecture 22	Isometric Views(hands-on-practice on 3D drawings)	T-1	AV-14	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
Week 12	Lecture 23	Development of Surfaces (Introduction)	T-1 R-1 R-2 R-3	AV-35 AV-36 AV-37 AV-38	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (development methods)	T-1 R-1 R-2 R-3	AV-35 AV-36 AV-37 AV-38	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
	Lecture 24	Development of Surfaces (surface development of regular and truncated prisms and pyramids)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (AutoCAD commands for 3D- subtract)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (union)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects

Week 12	Lecture 24	Development of Surfaces (orbit)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (and visual styles)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
Week 13	Lecture 25	Development of Surfaces (surface development of regular and truncated prisms and pyramids)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (AutoCAD commands for 3D- subtract)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (union)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (orbit)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (and visual styles)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects

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Week 13	Lecture 26	Development of Surfaces (surface development of regular and truncated prisms and pyramids)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (AutoCAD commands for 3D- subtract)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (union)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (orbit)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
		Development of Surfaces (and visual styles)	T-1 R-1 R-2 R-3	AV-14 AV-39 AV-40	Introduction to conceptual frameworks of the development of surfaces with their importance, use, and practice	Student must use the methods to draw the different surfaces development following the standard rules and methods	Presentations, white board demonstrations, AV aids	3D views of real-life objects
Week 14	Lecture 27	Development of Surfaces (hands-on-practice on 3D drawings)	T-1	AV-13 AV-14 AV-39 AV-40	To introduce the several commands to create, modify and operate the 3D shapes on AutoCAD	Student must be familiar with the several commands to setup the software interface for the creation of 3D views and their allied operations	Practice on AutoCAD software	Software
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			

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Week 15	Lecture 30			Spill Over			
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Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Project - Design project	Yes	25	CO1, CO2, CO3, CO4, CO5, CO6
Test 3	NO	37.5	CO5, CO6
Test 2	NO	37.5	CO3, CO4
Test 1	NO	37.5	CO1, CO2

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project - Design project	To evaluate the students' practical understanding/know-how of using an AutoCAD tool for 2D and 3D modeling skills	AutoCAD environment with potentials/ benefits, practice of orthographic (2D), sectioning, 3D modeling based design problems	Individual	Offline	30	2 / 10
Test 1	To evaluate the students' understanding of fundamental concepts of engineering drawing with projection of points and lines	Introduction to engineering drawing, projection of points and lines	Individual	Offline	30	5 / 6
Test 2	To evaluate the students' understanding of drawing of orthographic and sectional views	Orthographic and sectional views	Individual	Offline	30	9 / 10

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Test 3	To evaluate the students' understanding of drawing of isometric views and development of surfaces	Isometric views and development of surfaces	Individual	Offline	30	13 / 14
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List of suggested topics for term paper[at least 15] (Student to spend about 15 hrs on any one specified term paper)

Sr. No.	Topic
1	Design and development of following reduced scale products: 1) Mobile 2) Glass holder 3) Wall brackets 4) Electric fan 5) CPU fan 6) Water bottle 7) Mobile holder 8) Shoe base 9) Any toys shape 10) Lunch box 11) Executive Chair 12) Car wheels 13) Spring 14) Watch dial 15) Pipe connector

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	Practice- Line types and Dimensioning methods	Problem Solving
Tutorial2	Practice- Gothic letter writing	Problem Solving
Tutorial3	Practice- plain and diagonal scales	Problem Solving
Tutorial4	Practice- plain and diagonal scales	Problem Solving
Tutorial5	Practice of AutoCAD commands- units, limits, navigation, OSNAP, ortho, UCS, F-keys	Problem Solving
Tutorial6	Practice- orthographic projections of points and lines	Problem Solving

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Tutorial7	Practice- orthographic projections of points and lines	Problem Solving
Tutorial8	Practice of AutoCAD commands- line, circle, arc, polyline, rectangle, polygon, ellipse, dimensioning and style, hands-on-practice on AutoCAD	Problem Solving
Tutorial9	CA-I	Test
Tutorial10	Practice- Orthographic projections	Problem Solving
Tutorial11	Practice- Orthographic projections	Problem Solving
Tutorial12	Practice- Orthographic projections	Problem Solving
Tutorial13	Practice- Orthographic projections	Problem Solving
Tutorial14	Practice of AutoCAD commands as per the syllabus of unit-3	Problem Solving
After Mid-Term		
Tutorial15	Practice- Sectional Views	Problem Solving
Tutorial16	Practice- Sectional Views	Problem Solving
Tutorial17	Practice- Sectional Views	Problem Solving
Tutorial18	CA-II	Test
Tutorial19	Practice- Isometric views of Prisms	Problem Solving
Tutorial20	Practice- Isometric views of Pyramids	Problem Solving
Tutorial21	Practice- Isometric views of one object on other	Problem Solving
Tutorial22	Practice of AutoCAD commands as per the syllabus of unit-5	Problem Solving
Tutorial23	CA-III	Test
Tutorial24	Practice- Surface development of prisms	Problem Solving
Tutorial25	Practice- Surface development of pyramids	Problem Solving
Tutorial26	Practice- AutoCAD commands for 3D	Problem Solving
Tutorial27	Practice- Surface development of truncated objects	Problem Solving
Tutorial28	CA-IV	Test

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